. SET- BETA

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Department of Computer Science

CSC 2210 Object Oriented Analysis and Design (Section: ALL)

Fall 2024-25

Mid Term Examination

Time: 2 hours

Total Marks: 80

Moderator: Rahul Biswas

General Instructions:

1. **Use the OMR sheet for PART A and B. Write the SET NAME (BETA) in the OMR Sheet.**
2. **For parts C & D, use the given space provided in this question paper.** You may use **PENCIL** to draw the diagrams.
3. Return the question paper and OMR sheet at the end of the examination. Marks on the right margin indicate full marks.

Specific Instructions:

1. **All the questions are based on OBE program outcomes.**
2. **Attempting Question number 43 is MANDATORY, it will be mapped to CO2.**

Name: [redacted]			Invigilator's Signature
ID: [redacted]	Permit: [redacted]	Section: [redacted]	

Part – A (Answer All) **Multiple Choice Questions** **[30 x 1 = 30 Marks]**

Next 5 questions refer to the **Class Diagram Case Study** of Question No. 43:

1. The relationship between **Member** and **Commodities** is:
   (A) Dependency (B) Generalization
   (C) Aggregation (D) None
2. The relationship between **Commodity** and **Service** is:
   (A) Aggregation (B) Composition
   (C) Reflexive Association (D) Generalization
3. The relationship between **Event** and **Segment** is:
   (A) Generalization (B) Composition
   (C) Reflexive Association (D) Aggregation
4. The 'refer' relationship of **Member** is:
   (A) Aggregation (B) Generalization
   (C) Reflexive Association (D) Composition
5. The relationship between **Member** and **Service** is:
   (A) Association (B) Generalization
   (C) Aggregation (D) Reflexive Association
6. Which Function calls among objects is represented by:
   (A) State Machine (B) Note
   (C) Interaction (D) Component
7. What does reflexive association mean?
   (A) Association among objects of two different classes (B) Association among objects of the same class
   (C) Association among objects of two different classes using index (D) Association among different sub-classes of the same parent
8. How many objects of a class will be involved in an association relationship with the objects of another class is defined by its:
   (A) Association Role (B) Multiplicity
   (C) Association constraint (D) Qualified Association
9. If the data for all the objects of a class need to be updated with the same values, which type of attribute should be chosen?
   (A) Default value attribute (B) Derived attribute
   (C) Slash (D) Static attribute

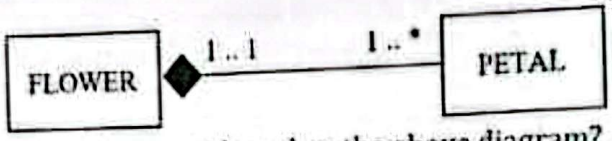
10. Which of the following addresses the runtime environment of the system?
    (A) Structural Things (B) Behavioral Things
    (C) Annotational Things (D) Grouping Things

11. A subclass can share the resources of its superclass. Which relationship in UML is used for this?
    (A) Generalization (B) Realization
    (C) Aggregation (D) Composition

12. If an attribute or an operation is only accessible within the class, what should be its visibility type?
    (A) Public (B) Protected
    (C) Private (D) Packaged

13. Alpha testing is expected at the end of the following phases of RUP:
    (A) Inception (B) Elaboration
    (C) Construction (D) Transition

Next 2 questions refer to the following diagram:



14. Which of the following statements is correct based on the above diagram?
    (A) A flower contains at least 1 petal. (B) A flower contains exactly 1 petal.
    (C) A flower contains may contain any number of petals or may not. (D) A flower contains minimum zero or maximum any number of petals.

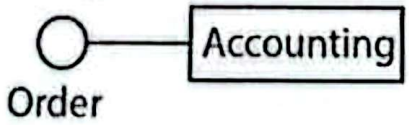
15. Which of the following statements is correct based on the above diagram?
    (A) Petal is not part of cycles. (B) A petal may or may not be part of a flower.
    (C) A petal may be a part of many flowers. (D) A petal must be a part of only one flower.

16. Class Diagram represents:
    (A) Static Use Case View (B) Static Design View
    (C) Dynamic Process View (D) Dynamic Interaction View

17. Which of the following is required for Objects to interact with each other?
    (A) Activity (B) Stereotype
    (C) Relationship (D) Note

18. A composition relationship must be a(n) _____ first.
    (A) Realization (B) Aggregation
    (C) Generalization (D) Dependency

19. What type of element is 'Accounting' in this design?



    (A) Note (B) Node
    (C) Artifact (D) Component

20. Which of the following contains **specialized** elements?
    (A) Parent Class (B) Super Class
    (C) Child Class (D) Interface

Match the notation of the **next 5 questions** with the items in the following list:

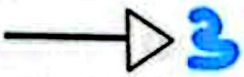
A. Interface B. Note C. Generalization D. Active Class E. Node

21. 1

22 4

23. 2

24. 5

25  3

26. Which of the following classes contains data about the relationship?
    (A) Component (B) Active Class
    (C) Interface Class (D) Association Class

Sec: ID:

Which of the following property is relevant to the Weighted Method per Class (WMC) metric?

A. Method bonding	B. Method complexity
C. Relationship	D. Inheritance

Which of the following property is relevant to the Coupling between Classes (CBC) metric?

A. Method bonding	B. Method complexity
C. Relationship	D. Attributes

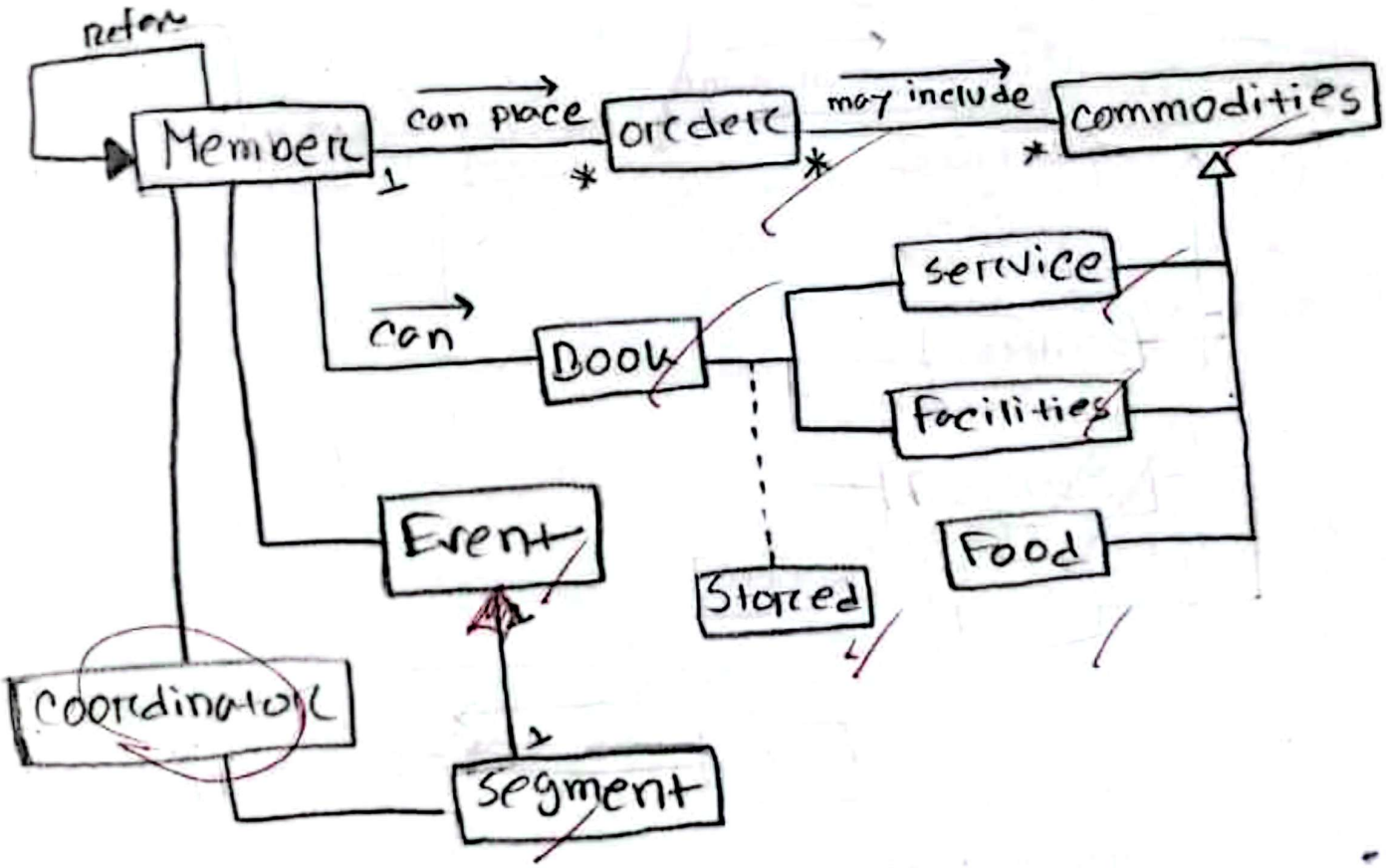
RUE/ FLASE: Darken circle 'A' for True; 'B' for False in the OMR: 11

15. LCOM value of a class is highly significant in identifying the fault-proneness of a class.
16. Cohesion is one of the main mechanisms for reuse.
17. If the LCOM value of a class is higher, the class might be considered to be separated into multiple classes.
18. Coupling between classes (CBC) of a system increases modularity.
19. DIT is significant in predicting defect-proneness of a class.
20. The root class(es) has the lowest DIT value.
21. Interface classes tend to have lower CBC values.
22. The higher NOC classes are more defect-prone.
23. Higher CBC value makes the class modification harder.
24. Inheritance increases coupling.
25. High cohesion is a highly desirable property for modularity.

Part – D (Answer All Questions)	Diagram	[1 x 15 = 15 Marks]
The answer to this part will be evaluated for the following OBE Course Outcome:		
Na[redacted]		[redacted]
CO2: *Determine the solution of a complex engineering problem using a UML diagram.*		**Total Marks:**
Object Oriented Concept (5)	Diagram Standard (5)	Correctness and Efficiency of the Design (5)	14
5	5	4	

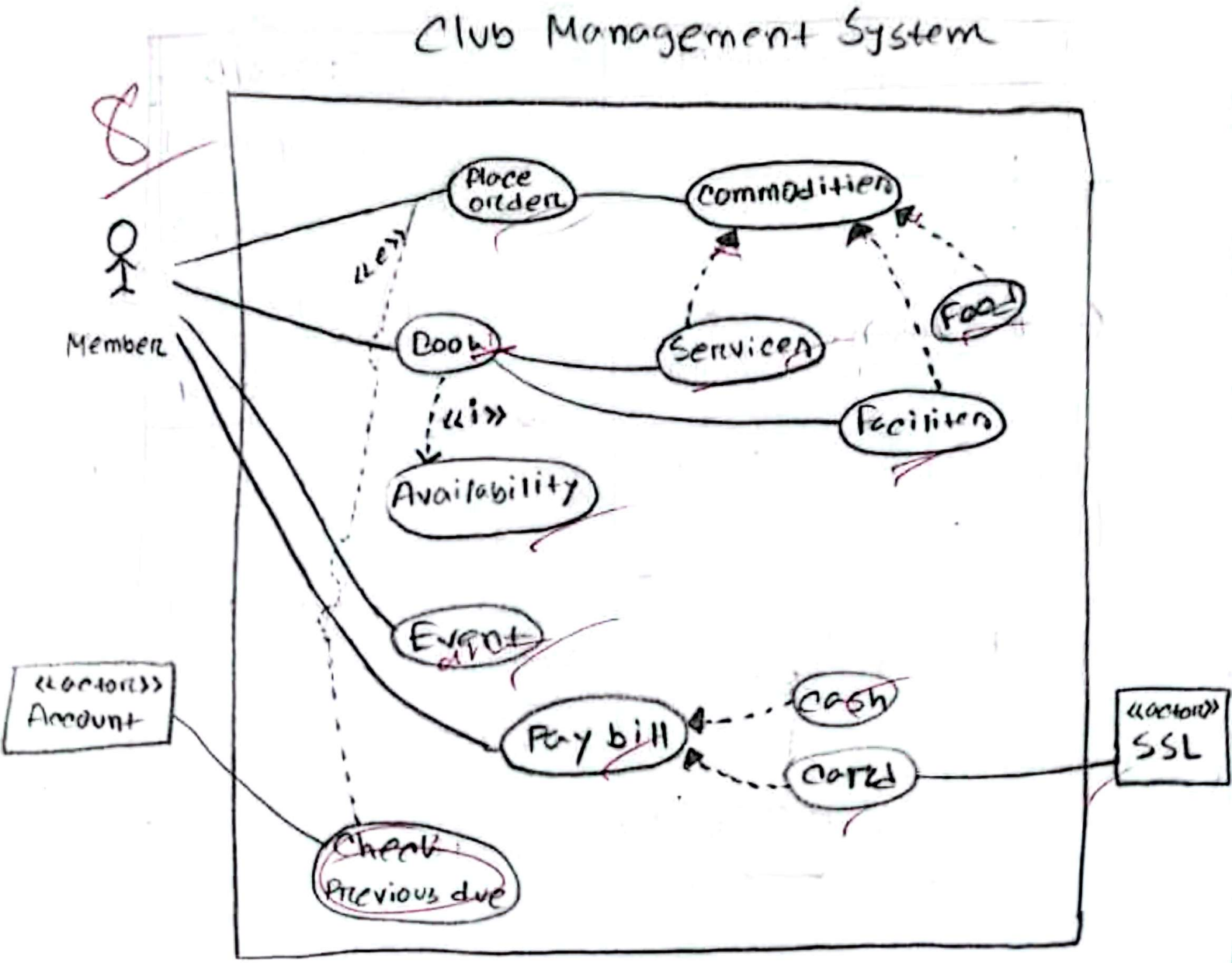
43. Draw a **Class Diagram** for the following order management system: **15 Marks**

In a club management system, members can place orders for commodities. There are various types of commodities, like, services, foods, facilities. One member can place many orders. One order is placed by exactly one member. One order may include many commodities, and one commodity may be included in many orders. Members can also book services and facilities. All booking information must be stored. The club also organizes events. Members are associated with the events. One event is made of at least one segment. One segment is part of only one specific event. Members are also assigned as the coordinator of the segments. One member may refer other members.



42. Draw a **Use Case** diagram for the following requirements **10 Marks**

In a club management system, members can place orders for commodities. There are various types of commodities, like, services, foods, facilities. Members can also book services and facilities. Booking depends on the availability checking done by the system. Members arrange events. Members can pay for their bills by case or card. Card payment is processed by a third-party agent called SSL. Member order placement depends on a clearance from the accounts department that the member does not have any previous due. For some special members, clearance is not required.

OR

Draw a CRC card for **'Member'** Class based on the information available in above cases of question no. 41, 42 and 43.